



DAVID H. MAISTER

University Health Services: Walk-In Clinic



Kathryn Angell stared out her office window at Holyoke Center in July 1980, oblivious to the bustle on Mount Auburn Street. In July 1979, shortly after receiving her master's degree in health policy and management from the Harvard School of Public Health, Angell had been hired as assistant director for ambulatory care at Harvard's University Health Services. A major objective of this new position was to reorganize the Walk-In Clinic—the topic of Angell's thesis.

Angell was the chief administrator of the clinic, responsible for its daily functioning, the organization of medical and support services, and overall planning. She was charged with improving the delivery of medical care through better coordination of services and implementation of new programs. Soon after assuming her duties, Angell implemented a triage system in the Walk-In Clinic, whereby arriving patients were screened by a triage coordinator to determine whether they should be treated by a nurse practitioner or a physician. A year later, Angell was ready to evaluate the clinic's performance.

University Health Services

University Health Services (UHS) offered medical care to Harvard University students, staff, faculty, and their dependents who elected health care plans that included UHS services. Because the system was prepaid for over 90% of the potential users, UHS operated primarily as a health maintenance organization.

The medical services offered included surgical and 24-hour emergency facilities, an inpatient infirmary, four outpatient clinics (including the Walk-In and three primary care clinics associated with specific Harvard professional schools), mental health services, laboratory and X-ray facilities, and a variety of other specialized services. Patients could choose a personal physician, who could be seen by appointment in the physician's office and who would, if necessary, refer the patient to a specialist. The outpatient clinics treated acute ailments and emergencies.

UHS was budgeted approximately \$10 million to meet its total health care expenses for the 1979–1980 fiscal year (**Exhibit 1**). Of the total, the Walk-In Clinic accounted for approximately 20%, including its share of professional and clerical labor, overhead, and supplies. Salaries for primary care physicians ranged from \$35,000 to \$55,000. Nurses were paid \$16,000 to \$26,000, depending on

Professor David H. Maister, Shauna Doyle and Rocco Pigneri prepared this case. It was revised by Professor Christopher W.L. Hart and Lucy N. Lytle. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

Copyright © 1981 President and Fellows of Harvard College. To order copies or request permission to reproduce materials, call 1-800-545-7685, write Harvard Business School Publishing, Boston, MA 02163, or go to <http://www.hbsp.harvard.edu>. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of Harvard Business School.

their experience. UHS incurred additional costs of 18.5% of salary in benefits for both physicians and nurses.

Physicians worked 40 hours a week for 46 weeks per year. Of the 40 hours, approximately 12 were spent in the Walk-In Clinic, 16 in meeting patients by appointment in the physician's office, 5 on duty at the UHS infirmary, and 7 on administrative and other matters. By well-established precedent, all UHS primary care physicians (general practitioners) were required to undertake duty in the Walk-In Clinic. Doctors who worked part-time were usually allocated a proportionate share of their time in the clinic. Included in the time for appointments (which were usually scheduled by the physician's secretary in half-hour intervals) were two half-hour periods per week known as "reserve time." These were periods when the doctor might ask patients to see her or him in the office, perhaps to check on the progress of treatment; only the physician could schedule appointments for reserve time. The physician could sometimes see up to four patients in one half-hour of reserve time. Most doctors preferred seeing patients in their office to Walk-In Clinic duty, partly because of the hectic pace of the Walk-In Clinic, but also because they preferred treating patients with whom they were familiar.

The Walk-In Clinic

The Walk-In Clinic at Holyoke Center provided the most comprehensive ambulatory care of the four UHS walk-in clinics. Patients with acute medical and surgical problems were served on a first-come, first-served basis Monday through Friday, 8 a.m. to 5:30 p.m. The clinic was also open on Saturdays from 8 a.m. to 12:45 p.m. Emergencies (of which there were few) were, of course, treated immediately. The clinic had 12 treatment rooms—4 for nurses and 8 for doctors. However, three of the doctors' rooms were permanently assigned to physicians as their UHS offices and were available for Walk-In Clinic use only when those physicians were scheduled for clinic duty.

In 1979, over 37,400 patients visited the Walk-In Clinic for treatment of problems ranging from colds, nausea, and respiratory illnesses to acute appendicitis and chest pains. Of the patients who visited the clinic, 67% were students, 23% were staff, and 10% were dependents and Medex and Medicare subscribers. One UHS study conducted over a three-week period in 1980 showed that an average of 143 patients were seen per day (**Exhibits 2 and 3**).

The Walk-In Clinic was staffed on the basis of past experience with peak periods of patient visits, which typically occurred between 10 a.m. and 4 p.m. (**Exhibit 4**). No criteria existed for establishing staffing levels; minor adjustments were made each year to increase coverage in busy periods. Kathryn Angell scheduled 22 physicians for specific times throughout the week, usually in blocks of three to four hours. Because of part-timers, administrative duties, and other factors, there were on average only 150 physician-hours available per week to the Walk-In Clinic. The clinic was also staffed by 2 registered nurses and by 11 nurse practitioners (registered nurses with additional medical training who could treat minor ailments without consulting a physician). A few nurse practitioners also treated patients by appointment. Nurses and nurse practitioners worked eight-hour shifts, including one hour for lunch. Nurse practitioners staffed the UHS emergency room at night and on weekends; they also performed a variety of semi-administrative duties, such as receiving test results over the phone. Approximately 45% of nurse practitioners' hours were available to treat patients in the Walk-In Clinic.

Pre-triage Organization

Before the triage system, a typical Walk-In Clinic patient provided basic identification information on a sequentially numbered sheet at the front desk, then took a seat in the waiting area. The receptionist next requested the patient's record from the Medical Records Department; retrieving it and sending it to the clinic took eight or nine minutes. The receptionist then brought the record to the medical desk, where a clerk checked to ensure that all recent lab reports were present and that the patient's address and phone number were current. When checking was completed, in about five minutes, the clerk placed the record and the numbered sheet in a pile ordered according to the arrival of patients. When a patient's record reached the top of the pile, he or she was seen by the first available nurse. If the problem was minor, the nurse provided the necessary treatment. If the patient still needed to see a doctor after 10 minutes of "diagnosis" with the nurse, he or she returned to the waiting area and the nurse put the medical record in a pile for the physicians, again according to the order of arrival. The patient was then seen by the first available physician.

Dissatisfaction with the Walk-In Clinic was widespread. The major complaint was the waiting time between sign-in and treatment. This period averaged 23 minutes, but as many as 22% of all patients had to wait more than 35 minutes for their first contact with a nurse. A November 1978 study of the clinic found that patients who requested specific physicians at sign-in waited an average of 40 minutes before seeing the desired staff member (these patients were not seen first by a nurse); this group accounted for approximately 19% of all patients. If a nurse had to refer a patient to a doctor, an average of 10 minutes elapsed between the end of the nurse visit and the meeting with the doctor. Some patients complained that the length of their wait was not related to the nature of the visit (such as a 55-minute wait for a prescription renewal). Other patients reportedly avoided visits to the clinic because of the expected wait. Patients viewed the Walk-In Clinic as cold, inefficient, and impersonal.

Members of the UHS administrative and medical staff also thought the clinic could function better. Sholem Postel, MD, deputy director and chief of professional services (physicians and nurses) at UHS, commented on the pre-triage system's problems:

All the nurses were involved in seeing all the patients initially. This created a bottleneck as each nurse independently decided the extent of care for a patient and then provided as much of that care as possible before, if necessary, having the patient wait to be seen by a physician. This led to inconsistency and too much variation in treatment, given the different skills and experience levels of individual nurses. Furthermore, though nurses saw all the patients who did not request a particular doctor, they treated only 40% of all patients definitively. The result: duplicated efforts (time, questions, and examinations) for those patients who were subsequently sent to an MD.

The Triage System

A triage system was introduced in September 1979 to overcome these problems. The system was defined as "the preliminary evaluation and referral of patients to the necessary health resource, based on decisions about the nature of the patients' problems and knowledge of the priorities and capabilities of the available health care resources." (Exhibit 5 compares the total processing times under the pre-triage and triage systems. Exhibit 6 presents the percentage of patients treated by the system's various providers.)

Under the triage system, the arriving patient filled out an Ambulatory Visit Form (AVF), which requested the reason for the visit as well as identification information (**Exhibit 7**). If the patient checked “emergency care,” the receptionist immediately notified a physician, nurse practitioner, or triage coordinator, who then assessed the patient’s condition. In most cases, however, the receptionist simply reviewed the AVF for completeness and requested the patient’s record from Medical Records. A clerk matched the record with the AVF, ensured that all personal information and prior tests were properly filed and updated in the record, and then placed the record chronologically in a “triage pile.”

One of two triage coordinators (highly experienced registered nurses) called for the patient initially. The triage coordinator visited briefly with the patient in a private room and then summarized the patient’s problem. If the coordinator decided the ailment warranted more immediate care, she put the patient ahead of others waiting to see a physician.

As one triage coordinator explained:

My duties are to determine the chief complaint of the patient and to triage him or her to a physician or nurse practitioner. I’ll spend three or four minutes per patient in an average encounter, and I rarely have to deviate from this—only when people are unable to clearly describe their symptoms or when they overestimate the severity of their illness. However, there is no time constraint in determining the status of a patient.

The triage coordinator did not treat the patient but determined, according to guidelines and her discretion, whether the patient needed to see a nurse practitioner or a physician in the Walk-In Clinic, or whether the problem could be better handled by an appointment or referral to another UHS service. Patients were triaged to a nurse practitioner if their ailments fell under one of 13 categories (**Exhibit 8**). All ailments outside these categories required the attention of a physician, unless the triage nurse felt a nurse practitioner could treat the problem. If a nurse practitioner treated a problem that was not included in the 13 categories, however, a physician’s authorization was required. This meant the nurse had to find a doctor who would sign the medical record. In some cases, the doctor chose to meet with or examine the patient before signing the medical record. Other doctors would sign without further examination. Expanding the guidelines beyond the 13 ailments would, by state law, require detailed treatment guides so that a nurse practitioner could treat the patient without consulting a physician. UHS planned such expansion in the near future, though it was not known how many patients this might affect.

After the visit with the triage coordinator, the patient returned to the waiting area while his or her record was placed chronologically in the nurse practitioner or the physician pile, unless more immediate care was deemed necessary. Physicians saw an average of 3.1 patients per hour, and nurse practitioners, about 1.8 patients per hour, though the numbers varied widely.

When the triage system was instituted, UHS administrators expected that the waiting time to see a triage coordinator would be about 15 minutes and that waiting time to see a nurse practitioner or a physician would be less than 10 minutes. A 1980 UHS study reported, however, that patients actually waited a mean of 19.7 minutes to see the coordinator and a mean of 18.6 minutes from the start of the visit with the coordinator to the start of the visit with a nurse practitioner or a physician. The average total waiting time was 37.5 minutes, including triage time (**Exhibit 9**). Angell commented:

When we introduced the triage system, we thought the nurse practitioners would accept more of the patient load and leave the physicians more time per patient. Unfortunately, it has not worked out that way. Among the reasons may be that the triage coordinators are sometimes classifying patients as MD/NP (physician/nurse practitioner) to maintain the flow

when they feel the practitioners are backed up. The physicians' share of patients thus gets increased in overload situations. We did not want to have MD/NP as a classification and have asked the triage coordinators to stop using it. When in doubt, they are to triage the patient to a nurse practitioner.

About 5% of the patients seen by a nurse practitioner were referred to a physician. If the additional treatment was expected to be brief, the patient remained in the nurse practitioner's room until the doctor arrived. If longer treatment by the physician was anticipated, the patient joined the MD waiting line, according to his or her AVF number. Thus, any other patient in the MD queue who had arrived before the referred patient would still be seen first. Although the mean times to be triaged to a doctor or nurse practitioner were about the same (approximately 19 minutes), the mean waiting time to see a physician was much longer (25.2 minutes) than was the mean waiting time to see a nurse practitioner (6.7 minutes).

One reason for this time difference may have been the percentage of patients who asked to see a specific doctor or nurse. After the triage system was instituted, this proportion increased to 24% of all patients, up from 19% previously. These patients still had to see a triage nurse first, who might try to dissuade them from waiting for a specific provider. If the patient chose to wait, however, he or she did not receive priority over patients who had arrived earlier. For almost one-third of the physicians, more than 40% of their clinic patients requested them (**Exhibit 10**).

Although the waiting time for triage was the same for the patients who asked for a specific provider and for those who did not, the former waited an average of 8.6 minutes longer to see a provider. But as Mary Dineen, nurse practitioner and supervisor of outpatient nursing, commented:

It seems doctors are allowed "walk-in appointments" with their own regular patients. Patients whose doctors have heavily booked appointment schedules become aware of the doctor's walk-in schedule and come to the Walk-In Clinic at prearranged times to meet. This may be a necessary evil to some degree, but today, for example, two of the five doctors on duty are 100% occupied with "walk-in appointments." This decreases our available MD resources by 40% for true walk-in patients today and fills up our waiting room.

Peter Zuromskis, a physician in the Walk-In Clinic, also suggested reasons for the misuse of the walk-in operation:

My evaluation of the dissatisfaction our patients have sometimes expressed with this system is that it represents an approach to acute ambulatory care quite different from that they have previously experienced. Patients understandably find appealing the nostalgic image of the general practitioner who knows his patients well and is able to provide advice and treatment of minor illnesses in his office with an apparent minimum of clerical encumbrances. This is clearly impossible in a clinic that provides the volume and variety of medical care services UHS offers to a large and heterogeneous population with a wide variety of diseases, from relatively minor complaints to major medical emergencies. Our aim is not and should not be to provide an atmosphere reminiscent of the country doctor's office, but rather to provide the best possible care to all our patients, particularly those whose medical needs are most urgent.

Although the new system had been receiving unsolicited praise, Angell knew it still had problems. She worried that by focusing on the "mechanics" of the problem she would overlook other dimensions. Some patient complaints still noted "excessive" waiting times and misunderstanding of the triage system, illustrated by the following opinion submitted to the UHS Patient Advocate:

In order to see a doctor about a very simple problem (a mild sore throat), I have seen a triage nurse (who stamped my form and passed me on) and a nurse practitioner (who looked, felt, and probed, but dared not offer an opinion). I am now 30 minutes into my visit, much handled, but not within sight of a doctor."

However, the medical, clerical, and administrative staff within the Walk-In Clinic felt the triage system was an improvement, though the clinic still wasn't as efficient as they would like it to be. As Warren Wacker, MD, the director of UHS, commented:

Right now, I'm satisfied with the results of the triage system, and I expect the system to be operating very well in another year. Of course, we'll have to resolve some sticky issues in the meantime. For instance, we need to expand the 13 nurse practitioner guidelines and further define the roles of nurse practitioner and physician within the Walk-In Clinic. Another item is, how do we educate students in the Walk-In Clinic concept? Expectations of traditional medicine don't fit with the walk-in concept.

Angell's Dilemma

Angell now had the task of sorting through a year's data, the concerns raised by several groups associated with the clinic, and her own observations. What changes needed to be made, if any? Were waiting times now acceptable? Angell knew that work was in progress to expand the 13 nurse practitioner guidelines, but would this be enough to solve any remaining problems? She wondered to what degree waiting time in a service facility affected one's perception of the quality of service provided.

Among her biggest concerns were the "walk-in appointments." She commented:

In the past, we have asked the doctors to refrain from encouraging their patients to meet them in the Walk-In Clinic. However, the practice continues. Some of the doctors feel that they want their patients to see only them. Part of this is for medical reasons (the doctors wish to check on their patients' progress) and part of it is a general philosophy that medical care involves more than just treatment and that personal relationships add to both the quality of health care and the patient's perception of good service. Many patients, perhaps appropriately, have the attitude of wanting to see "my doctor." Apart from the fact that you can never dictate to doctors, the UHS has always had a philosophy of not trying to tell physicians how to practice medicine.

Part of the problem is the general availability of appointment time. All our patients have the freedom to select their own "personal physician" from among any of our doctors. However, this often means that some are overloaded. Our overall staffing level at UHS is set approximately to provide one physician per 2,000 people covered by our various health plans. At the moment, the only way we try to limit the number of patients "assigned" to any given doctor is by pointing out to the potential patient the difficulty of getting an appointment with an overloaded physician, and this is generally only done if the patient asks about it. We do not know how many patients each doctor is seeing as the patients' personal physician, since this is an arrangement made by the doctor and the patient and not a formal assignment.

There are a number of potential alternatives for dealing with this problem. We could try to educate our patient public on the separate purposes and missions of doctor appointments and the Walk-In Clinic—try to get them to use each appropriately. We could ask the triage coordinators to be a little more aggressive in asking patients who request a specific physician whether they really need to see that person and suggest alternatives. Ultimately, we could establish a firm policy of not accepting specific physician requests in the Walk-In Clinic.

Angell had these questions and more to consider over the next two weeks. She would then share her findings and proposals with Ms. Dineen and Dr. Postel, who would have to agree on any necessary changes and help implement them.

Exhibit 1 Income and Expense Statistics for UHS Operations

	1979–1980	1978–1979	1979–1980
<i>Income</i>			
Student health fee	\$3,390,023	38.2%	34.4%
Student insurance	1,636,925	17.3	16.6
Harvard University Group			
Health Program	900,212	7.0	9.1
Payroll assessment	1,589,487	16.9	16.1
Care for Medicare	252,074	1.6	2.5
Radiation protection	435,603	3.8	4.4
Other services	<u>1,628,448</u>	<u>15.2</u>	<u>16.9</u>
Total	\$9,832,772	100.0%	100.0%
<i>Expenses</i>			
Salaries, wages, and benefits	\$5,223,685	53.5%	53.7%
Student insurance	1,636,925	17.2	16.8
Building operations and			
maintenance	388,870	4.3	4.0
Medical/dental supplies	278,987	2.3	2.8
Outside laboratories	176,309	2.2	1.8
Malpractice insurance	49,048	1.1	.5
All other	<u>1,967,436</u>	<u>19.4</u>	<u>20.5</u>
Total	\$9,721,260	100.0%	100.0%

Exhibit 2 Patient Visits per Day / Arrivals per Hour

Daily Average of Patient Visits	
Monday	163
Tuesday	151
Wednesday	136
Thursday	137
Friday	<u>128</u>
Average	143
Average Number of Patient Arrivals Per Hour	
8 – 9 a.m.	18.2
9 – 10 a.m.	17.6
10 – 11 a.m.	16.8
11 – 12 noon	15.2
12 – 1 p.m.	11.8
1 – 2 p.m.	16.9
2 – 3 p.m.	16.2
3 – 4 p.m.	15.9
4 – 5 p.m.	11.6
5 – 6 p.m.	2.8

Exhibit 3 Patient Distribution by Reason for Visit

Reason ^a	% of Total
Emergency	1.4%
Medical: initial visit ^b	41.3
Medical: return visit	11.3
Medical: specific provider	24.0
Surgical: initial visit ^b	0.1
Surgical: return visit	0.8
Lab result	2.0
Premarital test	0.4
Blood pressure	2.2
Prescription: confirmed diagnosis	0.8
Prescription refill	2.0
Administrative	1.0
Other	1.7
Unspecified	<u>11.0</u>
	100.0%

^aAs indicated by the patient^bInitial visit for the specific complaint, not patient's first visit *ever* to the Walk-In Clinic.**Exhibit 4** Medical Professional Scheduling: Walk-In Clinic

	Monday		Tuesday		Wednesday		Thursday		Friday	
	#	#	#	#	#	#	#	#	#	#
	MDs	NPs	MDs	NPs	MDs	NPs	MDs	NPs	MDs	NPs
8–9 a.m.	2	2	2	2	2	2	2	2	2	2
9–10 a.m.	2.5	4	3	4	2.5	4	2	4	2.5	4
10–11 a.m.	5	4	4	4	5	4	5	4	5	4
11–12 noon	3	4	3	4	3	4	3	4	4	4
12–1 p.m.	3	2.5	2	2.5	2.5	2.5	3	2.5	2.5	2.5
1–2 p.m.	3	2.5	3	2.5	3	2.5	2	2.5	3	2.5
2–3 p.m.	3	4	4	4	3	4	3	4	4	4
3–4 p.m.	4	4	4	4	4	4	4	4	4	4
4–5 p.m.	3	2.5	2	2.5	2	2.5	3.5	2.5	3	2.5
5–6 p.m. ^a	1	2	1	2	1	2	1	2	1	2

Note: MD = medical doctor. NP = nurse practitioner (statistics do not include triage nurses).

^aThe clinic admitted its last patient at 5:30 p.m. Staff were required to stay until 6 p.m.

Exhibit 5 Comparison of Total Processing Times, Pre-Triage and Triage (minutes)

	Pre-Triage	Triage
I. Initial Processing		
• Sign-in/AVF form	2	2
• Record retrieval	8.5	8.5
• Record check	5	5
• Wait for triage	NA	4.2
• Triage	NA	3.5
II. Patients Treated by Nurse Practitioner		
• Wait for NP	7.5–19.5	6.7
• Treatment by NP	32.8	32.8
III. Patients Treated by Physician		
• Wait for NP	7.5–19.5	NA
• Evaluation by NP	10	NA
• Wait for MD	10	25.2
• Treatment by MD	19.4	19.4
IV. Patients Treated by Specific Provider		
• Wait for specific MD or NP	24.5	33.8
• Treatment by specific MD or NP	19.4, 32.8	19.4, 32.8

December 27, 1996

Exhibit 6 Percentage of Patients Treated by Various Providers

	Pre-Triage	Triage
Patients treated by nurse practitioner	40%	28% ^a
Patients treated by physician	41%	48% ^b
Patients treated by specifically requested physician or nurse practitioner	19%	24%

^aTotal percentage of patients treated by nurse practitioner: 33%, including specifically requested nurse practitioners.^bTotal percentage of patients treated by physician: 67%, including specifically requested physicians.

Exhibit 7 Ambulatory Visit Form

No: 82336

UNIVERSITY HEALTH SERVICES : WALK-IN CLINIC
AMBULATORY VISIT FORM

FOR PATIENT USE: PLEASE FILL OUT THIS SECTION COMPLETELY		FOR WALK-IN PATIENTS ONLY	
TIME & DATE NAME: _____ PLEASE PRINT: _____ BIRTHDATE: _____ LOCAL ADDRESS: _____ PHONE DURING THE DAY: _____ LOCATION OF VISIT: ☐ Harvard ☐ Medical Area ☐ Other IS THIS YOUR FIRST VISIT TO A UHS FACILITY? YES ☐ NO ☐		The following information is designed to help us treat you promptly and efficiently. All information will be kept confidential. If you do not wish to complete the rest of the form, please check "personal" and you will be seen in turn. WHAT IS THE REASON FOR YOUR VISIT? PLEASE CHECK: ☐ I NEED EMERGENCY CARE. ☐ BLOOD PRESSURE CHECK ONLY PRESCRIPTION(S) ONLY ☐ DIAGNOSIS CONFIRMED-INSTRUCTED TO OBTAIN PRESCRIPTION ☐ PRESCRIPTION REFILL: _____ UNUSUAL OTHER: _____	
IF YOUR MEDICAL RECORD IS KEPT AT A UHS FACILITY OTHER THAN HOLYOKE CENTER, PLEASE CHECK HERE: BUSINESS ☐ LAW ☐ MEDICAL AREA ☐ STATUS ☐ M/R UNDERGRAD., CLASS _____ ☐ STUDENT DEPENDENT WITH UHS COVERAGE ☐ GRAD. SCHOOL (Name) _____ ☐ STUDENT DEPENDENT WITH UHS COVERAGE ☐ LESLEY COLLEGE _____ ☐ UNDER 14 YEARS OLD ☐ EPISCOPAL DIVINITY SCHOOL _____ ☐ HUGHP DEPENDENT ☐ STAFF WITH HARVARD BC/BS _____ ☐ MEDEX DEPENDENT ☐ STAFF WITH HARVARD UNIVERSITY _____ ☐ SUMMER SCHOOL: STUDENT _____ ☐ GROUP HEALTH PROGRAM (HUGHP) _____ ☐ FAC. DEPENDENT ☐ STAFF WITH NO HARVARD INSURANCE _____ ☐ NON-MEMBER OF HARVARD UNIVERSITY ☐ HARVARD MEDEX _____ ☐ OTHER _____ ☐ MEDICARE (ONLY) _____		GENERAL MEDICAL PROBLEM ☐ FIRST VISIT FOR THIS PROBLEM ☐ RETURN (REPEAT) VISIT FOR THIS PROBLEM ☐ TOLD TO SEE: _____ NURSE OR DOCTOR _____ GENERAL SURGICAL PROBLEM ☐ FIRST VISIT FOR THIS PROBLEM ☐ RETURN (REPEAT) VISIT FOR THIS PROBLEM ☐ TOLD TO SEE: _____ NURSE OR DOCTOR _____ LABORATORY PROCEDURES ONLY ☐ LAB RESULTS DESIRED ☐ PRENATAL TESTS DESIRED ☐ PREGNANCY TEST REQUISITION	
TRIAGE TIME _____ TIME PT. SEEN _____		FOR UHS USE ONLY TYPE OF CONTACT ☐ WALK-IN ☐ APPOINTMENT ☐ BROKEN APPOINTMENT ☐ CANCELLED BY UHS ☐ CANCELLED BY PATIENT ☐ LEFT BEFORE BEING SEEN ☐ RESERVE ☐ OTHER _____	
PROVIDER 1 NUMBER _____ NAME _____ PROVIDER 2 NUMBER _____ NAME _____		PROVIDER NO.: _____ 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____ 7 _____ 8 _____ 9 _____ 10 _____ 11 _____ 12 _____ 13 _____ 14 _____ 15 _____ 16 _____ 17 _____ 18 _____ 19 _____ 20 _____ 21 _____ 22 _____ 23 _____ 24 _____ 25 _____ 26 _____ 27 _____ 28 _____ 29 _____ 30 _____ 31 _____ 32 _____ 33 _____ 34 _____ 35 _____ 36 _____ 37 _____ 38 _____ 39 _____ 40 _____ 41 _____ 42 _____ 43 _____ 44 _____ 45 _____ 46 _____ 47 _____ 48 _____ 49 _____ 50 _____ 51 _____ 52 _____ 53 _____ 54 _____ 55 _____ 56 _____ 57 _____ 58 _____ 59 _____ 60 _____ 61 _____ 62 _____ 63 _____ 64 _____ 65 _____ 66 _____ 67 _____ 68 _____ 69 _____ 70 _____ 71 _____ 72 _____ 73 _____ 74 _____ 75 _____ 76 _____ 77 _____ 78 _____ 79 _____ 80 _____ 81 _____ 82 _____ 83 _____ 84 _____ 85 _____ 86 _____ 87 _____ 88 _____ 89 _____ 90 _____ 91 _____ 92 _____ 93 _____ 94 _____ 95 _____ 96 _____ 97 _____ 98 _____ 99 _____ 100 _____	
SERVICE ☐ MEDICAL ☐ SURGICAL ☐ EMERGENCY ☐ ALLERGY ☐ DENTAL ☐ DERMATOLOGY ☐ EAR, NOSE, & THROAT ☐ EYE ☐ GASTROENTEROLOGY ☐ GYNECOLOGY ☐ IMMUNIZATION ☐ MENTAL HEALTH ☐ NEUROLOGY ☐ NUTRITION ☐ OBSTETRICS ☐ ORTHOPEDICS ☐ PEDIATRICS ☐ PHYSICAL THERAPY ☐ UROLOGY ☐ OTHER _____		PROVIDER NO.: _____ 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____ 7 _____ 8 _____ 9 _____ 10 _____ 11 _____ 12 _____ 13 _____ 14 _____ 15 _____ 16 _____ 17 _____ 18 _____ 19 _____ 20 _____ 21 _____ 22 _____ 23 _____ 24 _____ 25 _____ 26 _____ 27 _____ 28 _____ 29 _____ 30 _____ 31 _____ 32 _____ 33 _____ 34 _____ 35 _____ 36 _____ 37 _____ 38 _____ 39 _____ 40 _____ 41 _____ 42 _____ 43 _____ 44 _____ 45 _____ 46 _____ 47 _____ 48 _____ 49 _____ 50 _____ 51 _____ 52 _____ 53 _____ 54 _____ 55 _____ 56 _____ 57 _____ 58 _____ 59 _____ 60 _____ 61 _____ 62 _____ 63 _____ 64 _____ 65 _____ 66 _____ 67 _____ 68 _____ 69 _____ 70 _____ 71 _____ 72 _____ 73 _____ 74 _____ 75 _____ 76 _____ 77 _____ 78 _____ 79 _____ 80 _____ 81 _____ 82 _____ 83 _____ 84 _____ 85 _____ 86 _____ 87 _____ 88 _____ 89 _____ 90 _____ 91 _____ 92 _____ 93 _____ 94 _____ 95 _____ 96 _____ 97 _____ 98 _____ 99 _____ 100 _____	
INITIAL VISIT FOR THIS PROBLEM RETURN VISIT		PROVIDER NO.: _____ 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____ 7 _____ 8 _____ 9 _____ 10 _____ 11 _____ 12 _____ 13 _____ 14 _____ 15 _____ 16 _____ 17 _____ 18 _____ 19 _____ 20 _____ 21 _____ 22 _____ 23 _____ 24 _____ 25 _____ 26 _____ 27 _____ 28 _____ 29 _____ 30 _____ 31 _____ 32 _____ 33 _____ 34 _____ 35 _____ 36 _____ 37 _____ 38 _____ 39 _____ 40 _____ 41 _____ 42 _____ 43 _____ 44 _____ 45 _____ 46 _____ 47 _____ 48 _____ 49 _____ 50 _____ 51 _____ 52 _____ 53 _____ 54 _____ 55 _____ 56 _____ 57 _____ 58 _____ 59 _____ 60 _____ 61 _____ 62 _____ 63 _____ 64 _____ 65 _____ 66 _____ 67 _____ 68 _____ 69 _____ 70 _____ 71 _____ 72 _____ 73 _____ 74 _____ 75 _____ 76 _____ 77 _____ 78 _____ 79 _____ 80 _____ 81 _____ 82 _____ 83 _____ 84 _____ 85 _____ 86 _____ 87 _____ 88 _____ 89 _____ 90 _____ 91 _____ 92 _____ 93 _____ 94 _____ 95 _____ 96 _____ 97 _____ 98 _____ 99 _____ 100 _____	

Exhibit 8 Categories Treatable by Nurse Practitioners

1. Acute viral respiratory illness (primarily colds)
 2. Amenorrhea (missed menstruation)
 3. Cerumen (wax in ears)
 4. Enterobiasis (pinworms)
 5. Lower urinary tract infection (females)
 6. Mononucleosis
 7. Nausea, vomiting, diarrhea
 8. Pediculosis capitis (lice)
 9. Pediculosis pubis (lice)
 10. Pharyngitis (sore throat)
 11. Rubella (German measles)
 12. Seasonal rhinitis (hay fever)
 13. Vaginitis (vaginal infection)
-

Exhibit 9 Percentage of Patients Waiting, by Time Waited

Interval (minutes)	To be Triage ^a	Waiting Time to be Seen ^b	Total Waiting Time
0–4	1%	24%	0%
5–9	8	14	3
10–14	24	12	7
15–19	25	11	10
20–24	19	9	10
25–29	11	8	14
30–34	6	8	11
35–39	2	5	10
40–44	1	4	8
45–49	1	3	6
50–54	1	2	7
55+	<u>1</u>	<u>0</u>	<u>14</u>
	100%	100%	100%
Average	19.7	18.6	37.5

^aMeasured from time patient received AVF form^bMeasured from time triaging begins to time patient meets physician or nurse practitioner**Exhibit 10** Summary of Patients Seen and Waiting Time to First Available Appointment, by Physician (over three-week period)

Physician ^a	Total Number of Patients Seen	Number of Patients Who Asked to See Specific MD (%)	Total Number of Hours	Number of Patients per Hour	Calendar Days to First Available Appointment
Zuromskis	113	33 (29.2)	35	3.14	9
Bogota	50	23 (46.0)	17	2.94	24
Wellington	89	-- --	18	4.94	5
Byrd	76	26 (34.2)	33	2.30	15
Recife	78	48 (61.5)	24	3.25	25
Brunei	113	45 (39.8)	36	3.14	17
Lobito	28	10 (35.7)	6	4.67	21
Santiago	91	43 (47.3)	29	3.24	3
Hobart	59	27 (45.8)	24	2.46	28
Seoul	90	34 (37.8)	28	3.21	5
Kingston	113	26 (23.0)	25	4.52	7
Java	78	16 (20.5)	27	2.89	13
Rome	74	32 (43.2)	19	3.89	7
Ottawa	82	31 (37.8)	26	3.15	5
Caracus	53	17 (32.1)	18	2.94	7
Manila	25	18 (72.0)	9	2.78	23
Durban	48	41 (85.4)	18	2.67	29
Luanda	61	5 (8.2)	21	2.90	8
Papua	34	-- --	9	3.78	--
Glasgow	35	2 --	9	3.89	12
Cristobal	33	3 (9.1)	19.5	1.69	2
Aukland	<u>16</u>	<u>1 (6.3)</u>	<u>12.5</u>	<u>1.28</u>	<u>--</u>
	1,439	481 (3.4)	463	3.10	12

^aSome names in this exhibit have been disguised.